



In silico prediction of protein–protein interactions

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Introduction

Protein-protein docking is the process of predicting how two proteins come together to form a complex. Determining these complexes experimentally is often more difficult than studying single proteins, so computational methods are very valuable[4]. Docking not only helps understand protein interactions but also provides a basis for drug design. Although there are some challenges, such as dealing with protein flexibility and low-resolution models, advances in computing power and structural knowledge have greatly improved docking methods[7]. At its foundation, docking still relies on the principle that proteins recognize and bind each other through complementary shapes and chemical features [7].

Materials and Methods

Software and Tools

We used **FRDOCK**(Fast ROTational DOCKing) [3] to perform the docking simulations. The **CAPRI table** [2] provided us with a list of 69 targets.

CAPRI (*Critical Assessment of Predicted Interactions*) is an international, community-wide blind assessment in which docking / PPI-prediction methods are tested against previously unpublished crystal or NMR complex structures[1]. CAPRI has established standard metrics for assessing how closely a predicted protein–protein complex matches the experimental reference structure[1], shown in the table below (Table1).

The metric we used is the L_{RMSD} , computed as the root-mean-square deviation of the ligand backbone heavy atoms (N, CA, C, O) between the native complex and each docked pose.

Table 1: CAPRI quality thresholds based on L_{RMSD} .

Quality level	Threshold
High quality	$L_{\text{RMSD}} \leq 1.0 \text{ \AA}$
Medium quality	$L_{\text{RMSD}} \leq 5.0 \text{ \AA}$
Acceptable quality	$L_{\text{RMSD}} \leq 10.0 \text{ \AA}$
Incorrect	$L_{\text{RMSD}} > 10.0 \text{ \AA}$

PyMOL[5] was applied for structure visualization and inspection. **Python** [6] was used for scripting and automation.

Docking Procedure

We attempted generating protein–protein complexes for 69 CAPRI targets (7 targets were duplicates though) using FRODOCK[3]. Structures were downloaded from the PDB using the CAPRI target list (column *PDB Final*)[2]. For each complex, we removed all non-protein parts and each complex was then split into two PDB files: receptor and ligand, with the receptor defined as the bigger partner and the ligand defined as the smaller partner. The receptor and ligand were chosen manually, by opening each PDB structure and inspecting the chains, as well as using PyMOL for a better inspection. For each receptor–ligand pair, we ran FRODOCK’s search and clustered the resulting poses. We evaluated only the highest-ranked cluster centers.

Predictions were assessed against the corresponding experimental complexes using the CAPRI L_{RMSD} , computed with Python scripting.

Pipeline overview is shown in figure 1. Below are detailed steps we used, while the commands can be found in the Appendix:

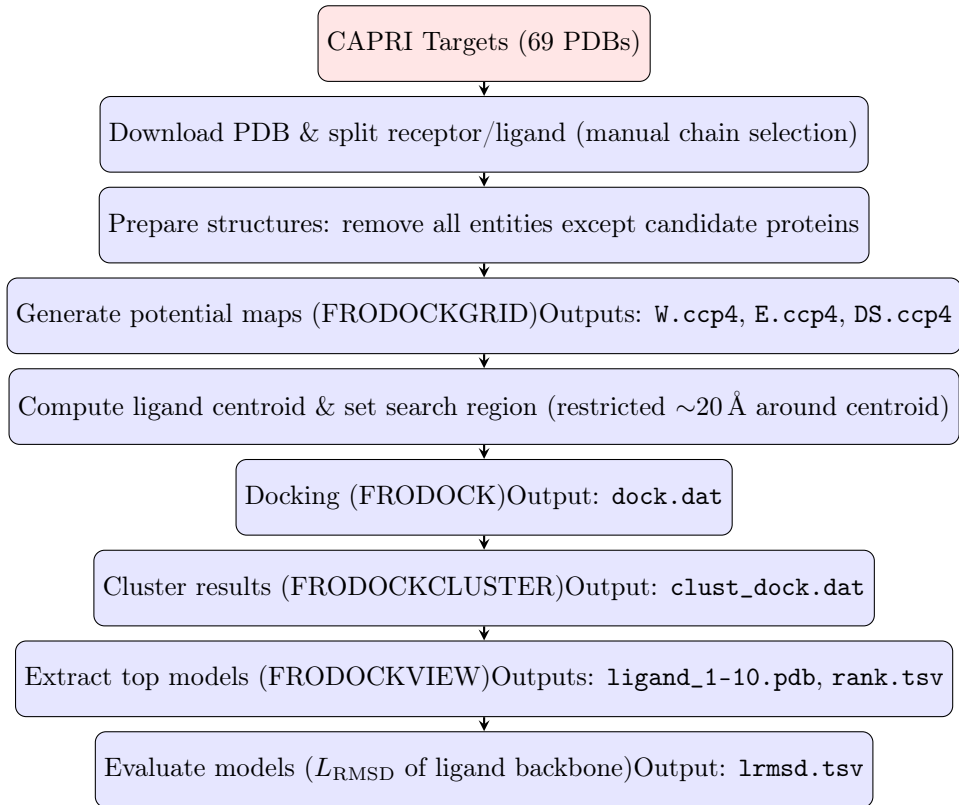


Figure 1: Docking workflow for CAPRI targets.

Results

Table 2: Summary of CAPRI docking targets with receptor/ligand chains, poses, and best L-RMSD values.

PDB ID	Receptor Chain(s)	Ligand Chain(s)	Pose	Best L_RMSD (Å)
PDBID	ReceptorChain(s)	LigandChain(s)	Pose	BestL_RMSD(Å)
1SYX	A	D	9	17.534
2B8W	A	B	1	1.419
2J59	A	B	17	21.959
2HQS	A	C	13	32.547
2O25	A	C	17	23.525
2ONI	Single chain	-	-	-
2VDU	Structure missing	-	-	-
2R2O	Single chain	-	-	-
2JPH	Single chain	-	-	-
3BX1	A	C	2	2.520
2W5F	A	C too small	-	-
2W83	A	C	20	13.653
3FM8	C	A	1	2.781
3E8L	A	C	2	17.174
2WPT	A	B	1	1.337
2WQH	A	A	16	29.092
3Q87	B	A	1	1.685
3U43	B	A	1	2.466
4I57	Structure missing	-	-	-
3R2X	A	B	1	1.170
4BXG	Structure missing	-	-	-
4JW2	B	A	1	0.967
4JW3	A	B	1	3.429
4EEF	A	B	1	0.854
4AK2	Single chain	-	-	-
4G9S	Single chain	-	-	-
3ZIN	A	B	1	1.286
3ZIO	A	B	1	1.266
1KKL	A	H	17	11.783
1KEN	A	H	19	5.298
1KXT	A	B	13	9.324
1KXV	A	C	2	1.593
1KXQ	A	E	7	79.755
1L0X	A	B	16	19.497
1NPE	B	A	1	3.583
1URZ	A	B	2	35.167
1OHZ	A	B	1	1.352
1YNT	F	B	9	20.046
1S70	A	B	1	1.379
1V74	A	B	1	1.158
1TA3	B	A	8	2.093
1TE1	B	A	1	2.901
1T6G	C	A	2	3.151
2B3T	A	B	4	4.953
1ZHI	A	B	5	3.117
3ZIP	A	B	8	1.712

Continued on next page

PDB ID	Receptor Chain(s)	Ligand Chain(s)	Pose	Best L_RMSD (Å)
3ZIQ	A	C	13	1.641
3ZIR	A	B	5	1.154
4ZOU	A	A	18	30.327
4NL8	A	C	5	4.973
4N7H	A	B	6	3.009
4Q34	A	A	13	40.765
4R8P	A	B	1	1.201
4XL5	B	A	1	1.689
4UEM	A	B	13	6.498
4UEL	A	B	1	1.293
4UF6	A	B	1	1.218
4UF5	A	B	1	2.437
5A4P	Single chain	-	-	-
4UHP	C	D	1	1.597
4QKO	B	A	1	1.627
4RT6	A	B	1	1.023

Docking was successfully completed for **52 CAPRI targets** (**7** were duplicates, which we skipped). **9** targets could not be processed because the PDB entry was missing or contained only a single chain, and **1** target could not be completed due to excessive runtime. The results are summarized in Table 2 and illustrated in Figure 2.

Several cases show **very low L-RMSD** values (below 2 Å), including **4EEF** (0.854 Å), **4JW2** (0.967 Å), **4RT6** (1.023 Å), **3ZIR** (1.154 Å), **1V74** (1.158 Å), **3R2X** (1.170 Å), **4R8P** (1.201 Å), **4UF6** (1.218 Å), **4UEL** (1.293 Å), **3ZIO** (1.266 Å), **3ZIN** (1.286 Å), **1OHZ** (1.352 Å), **1S70** (1.379 Å), **2WPT** (1.337 Å), **2B8W** (1.419 Å), **1KXV** (1.593 Å), **4UHP** (1.597 Å), **4QKO** (1.627 Å), **3ZIQ** (1.641 Å), **3Q87** (1.685 Å), **4XL5** (1.689 Å), and **3ZIP** (1.712 Å).

Moderate results in the 2–5 Å range include **1TA3** (2.093 Å), **4UF5** (2.437 Å), **3U43** (2.466 Å), **3BX1** (2.520 Å), **3FM8** (2.781 Å), **1TE1** (2.901 Å), **4N7H** (3.009 Å), **1T6G** (3.151 Å), **1ZHI** (3.117 Å), **4JW3** (3.429 Å), **1NPE** (3.583 Å), **2B3T** (4.953 Å), and **4NL8** (4.973 Å).

Intermediate-to-poor predictions in the 5–10 Å range include **1KEN** (5.298 Å), **4UEM** (6.498 Å), and **1KXT** (9.324 Å).

Several targets exhibit **high errors** above 10 Å, such as **1KKL** (11.783 Å), **2W83** (13.653 Å), **1SYX** (17.534 Å), **3E8L** (17.174 Å), **1L0X** (19.497 Å), **1YNT** (20.046 Å), **2J59** (21.959 Å), **2O25** (23.525 Å), **2HQS** (32.547 Å), **1URZ** (35.167 Å), **4ZOU** (30.327 Å), **2WQH** (28.986 Å), **4Q34** (40.765 Å), and **1KXQ** (79.755 Å).

In a few instances, docking could not be performed because structures were missing or unsuitable: **2VDU** (structure missing), **2ONI**, **2R2O**, and **2JPH** (single-chain entries), and **2W5F** (ligand chain too small).

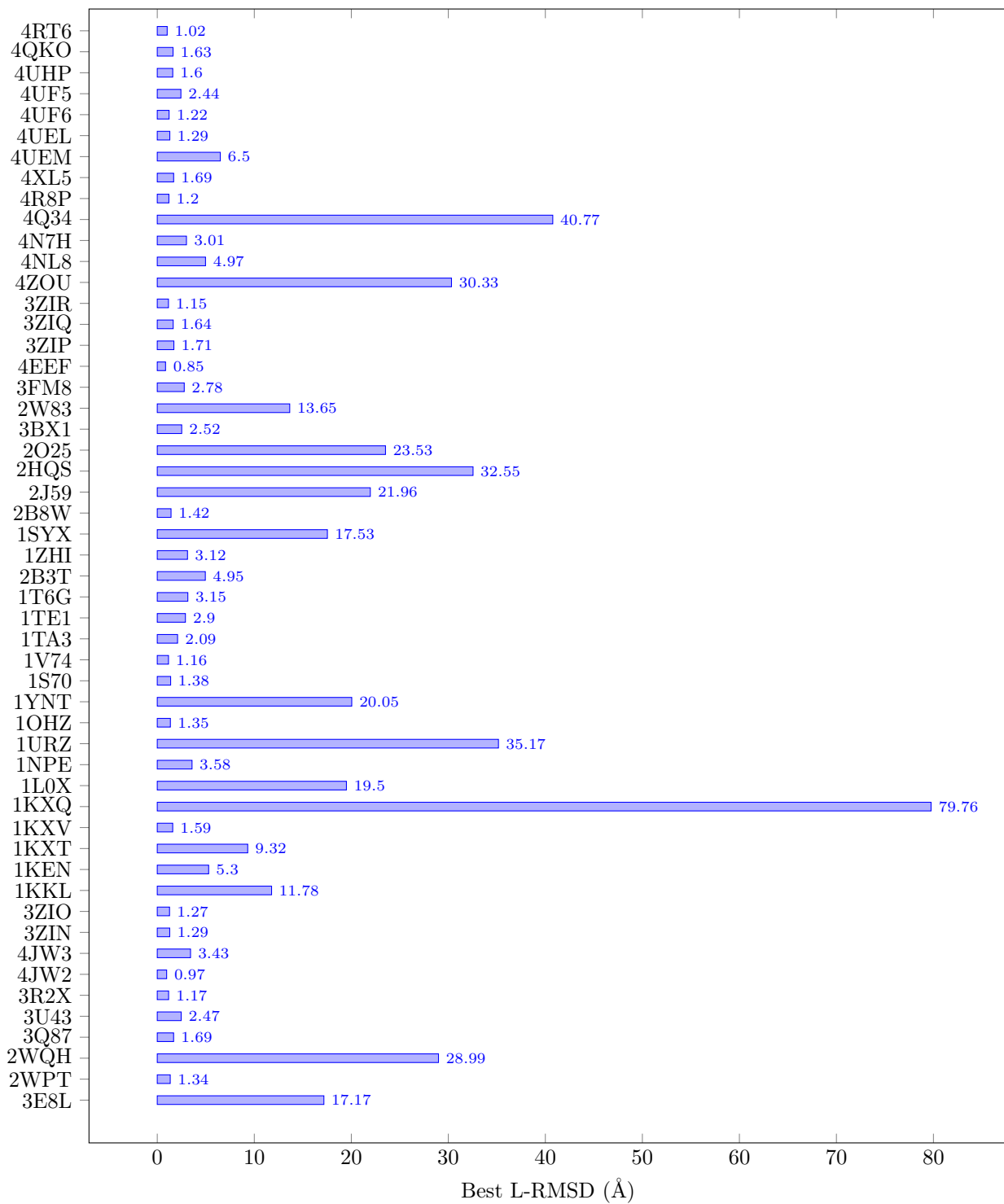


Figure 2: Best L-RMSD per target (top-ranked pose).

Discussion

Based on CAPRI criteria (Table1), we can conclude that our results show that the docking pipeline is able to generate near-native solutions in a meaningful number of cases, with several targets achieving L-RMSD below 2 Å (Figure4). These outcomes show that the method can capture correct binding orientations when receptor and ligand structures are well defined. Predictions in the range of 2–5 Å are still acceptable according to CAPRI, however, several targets produced very poor results with L-RMSD values above 10 Å, which indicates that the docking failed to recover the correct interaction. These failures may be explained by protein flexibility (unaccounted conformational changes), incomplete structural data (missing loops/side chains in the PDB), limited pose sampling (20 Å docking box), or potential errors in receptor/ligand chain selection done by us. Large multi-chain targets took much longer to dock (2–3 hours vs. ~30 minutes for smaller ones) but it doesn't have a statistically significant deviation in terms of the L-RMSD Values (L-RMSD > 10 Å) (Figure3).

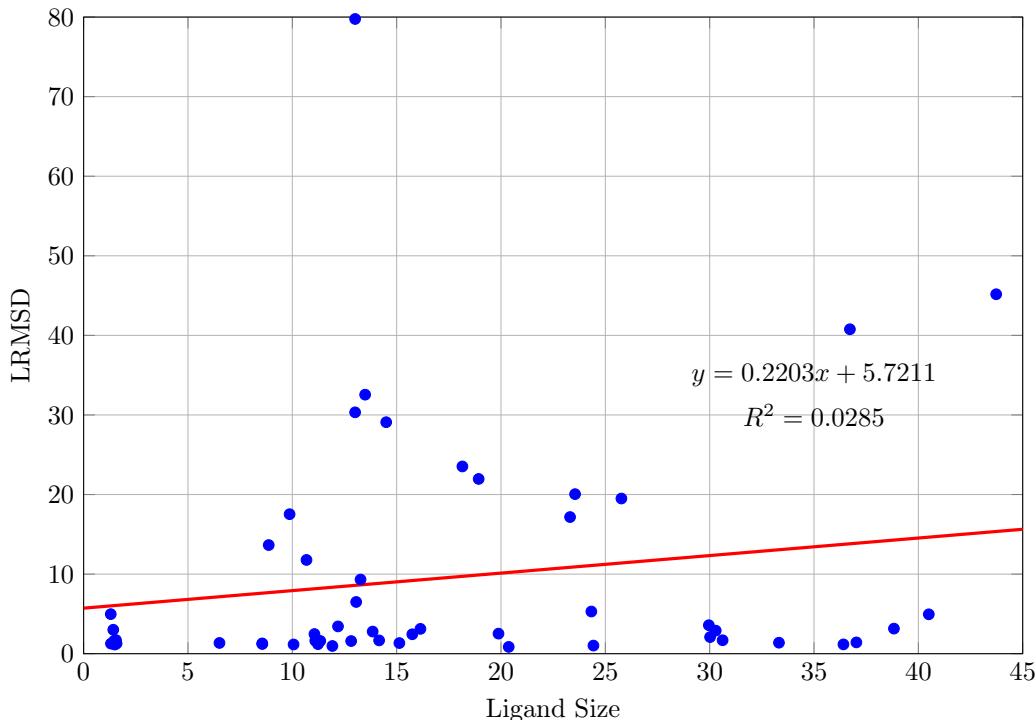


Figure 3: Distribution of ligand size versus docking quality across all 52 completed targets (top-ranked pose only), with regression line and corresponding R^2 value

This likely reflects higher complexity, a larger search space, and limits of rigid-body docking in handling chain flexibility. However, in our tests we docked only a single receptor–ligand chain pair from each multi-chain structure, so the poor performance is more likely due to interface difficulty and the limitations of rigid-body docking than to the total number of chains in the original PDB entries.

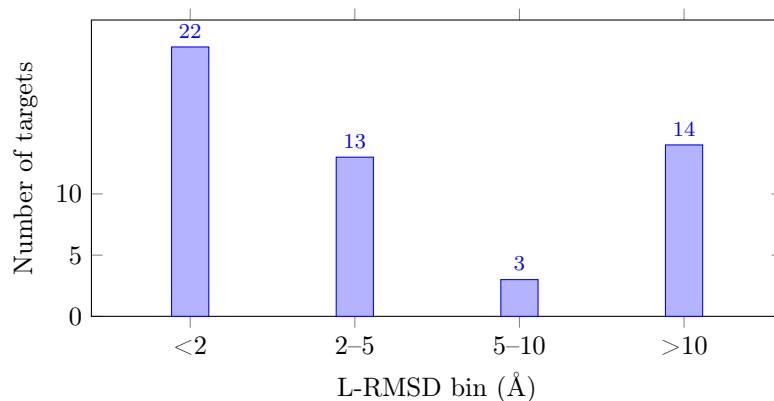


Figure 4: Distribution of docking quality across all 52 completed targets (top-ranked pose only).

In order to highlight the difference between successful and unsuccessful docking runs, we compared four representative targets. In this analysis, receptor–ligand interface residues were defined as those whose atoms lie within 4 Å of any ligand atom. These residues are shown as sticks (yellow/orange) in the figures, while the docked ligand is colored cyan and the experimental ligand magenta.

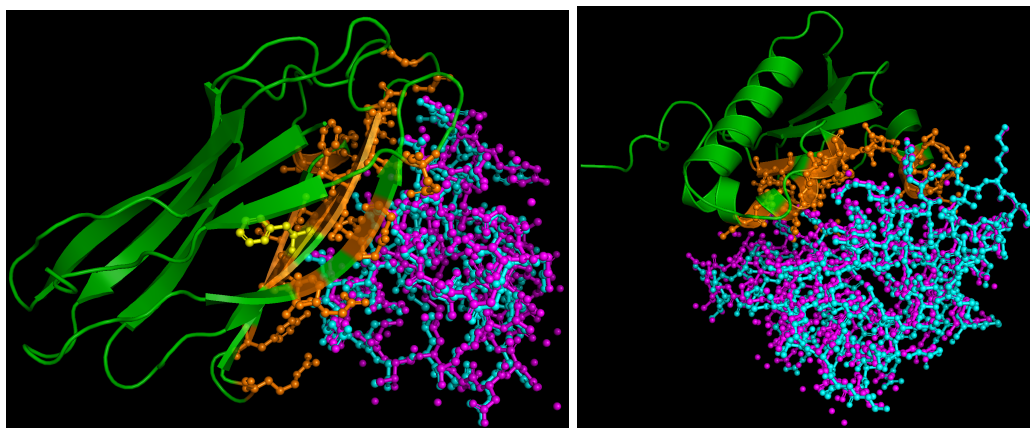


Figure 5: **Successful docking examples.** For 1OHZ (left, RMSD = 1.35 Å) and 1V74 (right, RMSD = 1.16 Å), the docked ligands align closely with the experimental ligands, producing a dense and continuous receptor–ligand interface.

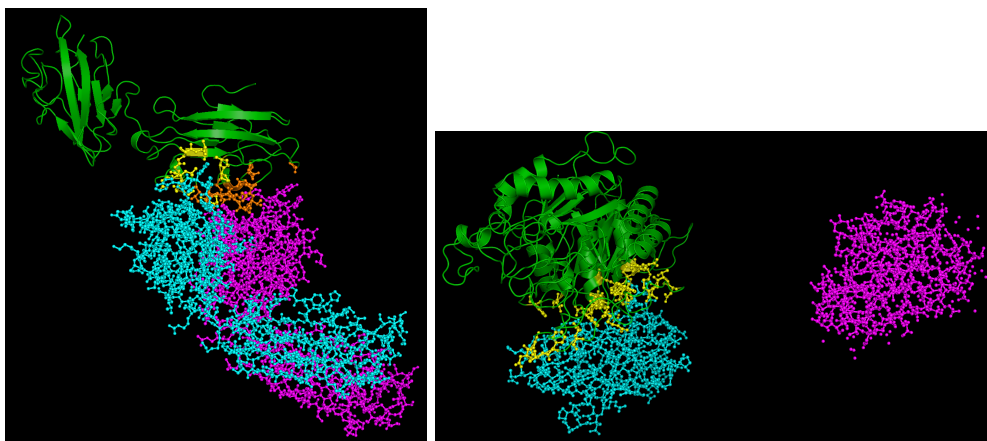


Figure 6: **Unsuccessful docking examples.** In 1YNT (left, RMSD = 20.05 Å) and 1KXQ (right, RMSD = 79.76 Å), the docked ligands are shifted far away from the experimental sites and a few scattered receptor residues lie within 4 Å, showing that the docking poorly reproduced the true binding modes.

Appendix

A Detailed pipeline with commands:

1. In order to split the structures into receptor and ligand chains, manually inspect the PDB files and directly write them into the code.
2. Download the PDB file and split receptor/ligand chains, filter the structures by removing all entities but the candidate proteins:

```
wget https://files.rcsb.org/download/<target_pdb>.pdb

# Receptor = manually selected receptor chain
awk '/^ATOM/ && !(substr($0,13,1)=="H" || toupper(substr($0,77,1))=="H") \
    && substr($0,22,1)=="<RECEPTOR_CHAIN>" \
    && (substr($0,17,1)==" " || substr($0,17,1)=="A")' \
"$TARGET.pdb" > "${target}_rec.pdb"

# Ligand = manually selected ligand chain
awk '/^ATOM/ && !(substr($0,13,1)=="H" || toupper(substr($0,77,1))=="H") \
    && substr($0,22,1)=="<LIGAND_CHAIN>" \
    && (substr($0,17,1)==" " || substr($0,17,1)=="A")' \
"$TARGET.pdb" > "${target}_lig.pdb"
```

3. Generate potential maps:

```
frodockgrid <target_pdb>_rec.pdb -o <target_pdb>_rec_W.ccp4
frodockgrid <target_pdb>_rec.pdb -o <target_pdb>_rec_E.ccp4 -m 1
frodockgrid <target_pdb>_rec.pdb -o <target_pdb>_rec_DS.ccp4 -m 3
frodockgrid <target_pdb>_lig.pdb -o <target_pdb>_lig_DS.ccp4 -m 3
```

(Outputs include steric, electrostatic, and desolvation maps)

4. Compute ligand centroid for a **restricted search window** (~20 Å) to speed up comp.time:


```

CENTER=$(
  awk '
    /^ATOM/ {
      x += substr($0,31,8) + 0
      y += substr($0,39,8) + 0
      z += substr($0,47,8) + 0
      n++
    }
    END {
      if (n > 0)
        printf "%.3f,%.3f,%.3f\n", x/n, y/n, z/n
    }
  ' <target_pdb>_lig.pdb
)

```

5. Run docking (maps + SOAP term, restricted around centroid):

```

frodock <target_pdb>_rec_ASA.pdb <target_pdb>_lig_ASA.pdb \
-w <target_pdb>_rec_W.ccp4 \
-e <target_pdb>_rec_E.ccp4 --th 10 \
-d <target_pdb>_rec_DS.ccp4,<target_pdb>_lig_DS.ccp4 \
-s /work/frodock3_linux64/bin/soap.bin \
--around "$CENTER" --td 20 \
-o <target_pdb>_dock.dat

```

6. Cluster docking results:

```

frodockcluster <target_pdb>_dock.dat <target_pdb>_lig.pdb --nc 100 -o <target_pdb>_clust.dat

```

7. Export top poses and generate a ranking table:

```

frodockview <target_pdb>_clust.dat -r 1-10
frodockview <target_pdb>_clust.dat -r 1-20 -p <target_pdb>_lig.pdb
frodockview <target_pdb>_clust.dat -r 1-50 > <target_pdb>_rank.tsv

```

8. Evaluate docking quality with Python and PyMOL (computing L-RMSDs):

This step was performed using a custom Python script (code attached), which generates **rmsd_top10.txt**.

References

- [1] Keeley W. Collins, Matthew M. Copeland, Guillaume Brysbaert, Shoshana J. Wodak, Alexandre M. J. J. Bonvin, Petras J. Kundrotas, Ilya A. Vakser, and Marc F. Lensink. Capri-q: The capri resource evaluating the quality of predicted structures of protein complexes. *Journal of Molecular Biology*, 436(17):168540, 2024.
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